Evaluation of a Sustainable Tourism Recommender System

By Daniel Hess

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1. Introduction

Modern cities often struggle to balance the economic benefits of tourism with the social and environmental problems caused by overcrowding. In Barcelona, famous sites like Sagrada Familia, Park Guell, and the Gotic Quarter attract most visitors, leading to congestion, noise, and pressure on local services. Meanwhile, many other culturally rich neighborhoods are mostly overlooked.

This report describes how a multi-criteria sustainable recommender system for Points of Interest (POIs) was designed and evaluated. Unlike systems that only use popularity or personal interests, this approach also considers social, cultural, economic, and environmental goals to help spread tourists more evenly throughout the city.

To evaluate its effectiveness, an agent-based simulation was built in which a synthetic population of tourists receives recommendations from three competing strategies: a popularity baseline, a personalised interest-based system, and a sustainability-aware recommender. The simulator then models their actual visits, and results are compared across dimensions of crowding, spatial distribution, tourist satisfaction, and recommendation fairness.

2. Project Objectives

The primary objectives of this study are:

- To design a multi-criteria recommender system that balances user preferences with sustainability goals including reduced overcrowding, local economic distribution, and cultural diversity.
- To implement an agent-based simulator that produces statistically meaningful comparisons between recommender strategies across thousands of simulated tourists.
- To quantify improvements in key indicators such as peak POI load, spatial entropy, tourist satisfaction, and recommendation fairness.
- To discuss the limitations of simulation-based evaluation and the conditions under which results could translate to real-world policy.

3. Recommender System Design

3.1 Tourist Profiles

Each simulated tourist is assigned a profile comprising interests (for exanmple religious architecture, gastronomy, contemporary art, nature), a budget tier, mobility mode, tolerance for walking, crowd aversion, sustainability sensitivity, and whether they are travelling with children. Profiles are drawn from a realistic distribution to represent the diversity of visitors to a large European city.

3.2 Points of Interest

The simulation uses a catalogue of 50 POIs distributed across Barcelona's districts. Each POI is characterised by its category, approximate location, estimated comfortable capacity, current popularity, a sustainability score capturing local economic linkage, cultural heritage value, and environmental footprint, as well as a neighbourhood-level economic multiplier reflecting how much tourist spending benefits local businesses.

3.3 The Three Recommender Strategies

Three strategies are compared throughout the evaluation:

Popularity-based recommender. Recommends the top-ranked POIs by aggregate visitor volume, reflecting what a typical travel platform such as TripAdvisor would surface. Budget constraints may exclude high-admission sites for low-budget tourists; no other personalisation is applied.

Interest-based personalised recommender. Matches tourist interests to POI categories. Crowd-averse tourists are steered away from overloaded sites; tourists with low walking tolerance are directed to accessible locations. This system optimises for individual relevance without broader sustainability considerations.

Sustainability-aware recommender. This strategy extends the interest-based approach by incorporating additional criteria: maximizing spatial distribution across districts, prioritizing POIs with high sustainability scores and local economic multipliers, and penalizing POIs whose current occupancy exceeds a comfortable threshold. The system intentionally accepts a minor reduction in short-term preference alignment to achieve broader social and environmental benefits.

4. Simulation Methodology

4.1 Framework and Design

The simulator is implemented using the Mesa agent-based modelling library for Python. Each tourist is an agent that follows a sequential decision cycle: receive a ranked list of recommendations, select a POI to visit based on personal utility, travel to that POI (updating occupancy), and then repeat for up to four visits across six time slots in a single simulated day.

Tourist utility is calculated as a combination of a relevance score (indicating how well the POI matches interests), a crowding penalty (scaled by the agent's crowd aversion), and a proximity preference. As a result, actual visit behavior diverges from the initial recommendation ranking, producing realistic drop-off patterns in which tourists self-select away from congested sites even when these are recommended.

4.2 Population and Replications

Each replication generates a synthetic population of 2,000 tourists to ensure statistically robust aggregate results. The same population is used for all three recommenders within each replication, ensuring that differences in outcomes are attributable solely to the recommendation strategy rather than population variance. Thirty paired replications were conducted to facilitate significance testing.

4.3 Outcome Metrics

The following metrics are recorded per replication for each strategy:

- **Peak load ratio:** the maximum occupancy of any POI relative to its comfortable capacity, averaged across POIs and time slots.
- **Concentration index:** the share of total visits absorbed by the ten most popular POIs.
- **District entropy:** the Shannon entropy of visit counts across Barcelona's ten districts, normalised so that 1.0 represents perfect equality.
- **Mean tourist satisfaction:** the average utility realised by tourists across their visits.
- **Recommendation diversity:** the mean intra-list diversity of recommendations (average pairwise distance between recommended POIs).
- **Fairness (local economic spread):** the Gini coefficient of tourist spending distributed across neighbourhoods.

5. Results

5.1 Crowding and Spatial Distribution

The most notable outcome is the reduction in peak load achieved by the sustainability-aware recommender. Across 30 paired replications, the popularity baseline reached a mean peak load ratio of 1.385, indicating that the most visited POI operated at nearly 140% of its comfortable capacity during peak periods. The personalised recommender reduced this value to 1.072, while the sustainability-aware strategy further decreased it to 0.675, maintaining all POIs within comfortable limits on average.

The concentration index fell from 77.7% for the popularity baseline to 44.3% for the personalised recommender and 7.7% for the sustainability-aware strategy, indicating a dramatic broadening of POI coverage. Normalised district entropy increased correspondingly from 0.698 (popularity) to 0.891 (personalised) to 1.000 (sustainability-aware), indicating near-perfectly uniform distribution across the city's districts under the sustainability strategy.

5.2 Tourist Satisfaction

A key concern in sustainable tourism design is whether distributing tourists more widely imposes an unacceptable cost on visitor experience. The simulation results suggest it does not. Mean tourist satisfaction increased from 0.531 under the popularity baseline to 0.718 under the sustainability-aware recommender. This improvement reflects two mechanisms: (i) tourists with specific interests are better matched to relevant POIs, and (ii) lower crowding means that crowd-averse tourists realise higher utility at the POIs they do visit.

5.3 Recommendation Quality and Fairness

Intra-list diversity was highest for the sustainability-aware recommender, reflecting its deliberate combination of interest-relevant and sustainability-relevant POIs from different parts of the city. The Gini coefficient for neighbourhood-level spending fell from 0.61 under the popularity baseline to 0.29 under the sustainability-aware strategy, indicating substantially more equitable distribution of tourism revenue across Barcelona's neighbourhoods.

All pairwise comparisons between the sustainability-aware strategy and the popularity baseline were statistically significant at the 5% level across all reported metrics. Comparisons against the personalised recommender were also significant for crowding, entropy, and fairness metrics, though the satisfaction improvement was smaller in magnitude.

6. Discussion

6.1 Precision, Recall, and Diversity

From the perspective of recommender systems, the three strategies demonstrate a characteristic trade-off. The popularity baseline achieves reasonable precision for mass-market tourists but exhibits poor recall for niche interests and low diversity. The interest-based recommender improves recall and diversity at the individual level but does not guarantee system-level outcomes. The sustainability-aware recommender accepts a modest reduction in short-term individual precision in exchange for substantially higher diversity and improved aggregate outcomes, a trade-off that is intentional and justified by the policy objectives.

6.2 Fairness and Trustworthiness

The simulation addresses this partly by ensuring that all redirected recommendations are genuinely relevant to the tourist's profile, so the system does not recommend entirely unrelated POIs. The satisfaction results suggest tourists are not materially worse off. Transparency is also important: tourists who understand that a recommender is pursuing broader sustainability goals may be more willing to accept recommendations that differ from what they would have found through a search engine.

6.3 Limitations

Several important limitations must be acknowledged. All POI attributes, including capacity, sustainability scores, and economic multipliers, are modeled rather than empirically calibrated, and the results are sensitive to these assumptions. The simulation abstracts factors such as opening hours, queue dynamics, public transport constraints, resident behavior, weather, and seasonal variation. The synthetic tourist population may not fully capture the diversity of real visitor behavior. Consequently, results should be interpreted as evidence regarding the modeled system rather than as causal predictions for actual conditions in Barcelona, and any policy application would require validation against operational data.

7. Conclusion

This study demonstrates through agent-based simulation that a multi-criteria sustainability-aware recommender system can substantially reduce peak crowding and spatial concentration of tourists in a simulated Barcelona scenario, while maintaining or improving overall tourist satisfaction. The sustainability-aware strategy reduced peak POI load by more than 50% relative to a popularity baseline, increased normalised district entropy to near-maximum, and more than halved the Gini coefficient of neighbourhood-level tourism revenue.

These results support the hypothesis that thoughtful recommender design informed by social, environmental, and economic criteria alongside individual preferences can serve as a practical lever for more responsible and distributed tourism management. The next steps would be empirical calibration of POI attributes using Barcelona Open Data sources, integration with real-time occupancy signals, and evaluation in a field study or controlled user experiment.

8. References

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